



Practical 1

Project Initiation, Requirement Analysis, Sitemap, Wireframe & GitHub Setup

StudentHub Portal — Student Management System

ITUE203 · Web Development Frameworks · Semester 3 (2026–27)

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Problem Definition

Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and a minimum of 10 pages. Produce a sitemap, a low-fidelity wireframe, a project folder structure, a README file, and a GitHub repository.

DELIVERABLES

- Functional requirements
- Sitemap
- Wireframe screenshots
- GitHub repository link
- README.md

TOOLS & TECHNOLOGY

- VS Code
- Draw.io / Figma
- Git
- GitHub

Implementation: 4 hrs | Engagement: 4 hrs

Requirement Analysis

User Roles

Visitor

Browses Home, About, Events, FAQ and Contact without logging in.

Student

Registers/logs in, views Dashboard, Events, and manages Profile.

Admin

Manages students, faculty, events, results and notices.

Key Modules (10 Pages)

Home

About

Register

Login

Dashboard

Events

Profile

Contact

Admin

FAQ

What is a URL?

A URL (Uniform Resource Locator) is the address used to locate a resource — such as a web page, image, or file — on the internet.

Scheme

`https://`

Protocol used to access the resource (e.g. HTTP, HTTPS).

Host

`www.studenthub.edu`

Domain name / server where the resource is located.

Port

`:443`

Network port the server listens on (often implied).

Path

`/dashboard`

Location of the specific resource on the server.

Query

`?user=25dce071`

Key–value parameters sent to the server.

Fragment

`#profile`

Points to a section within the resource itself.

How Does a Browser Process an HTML File?



This request–response and rendering cycle repeats for every linked page in the StudentHub site.

What is a Sitemap?

A sitemap is a structural diagram that lists every page of a website and shows how pages relate to and link with one another. It defines the site's information hierarchy before any code is written.

Plans navigation

Confirms every page is reachable from the menu.

Clarifies scope

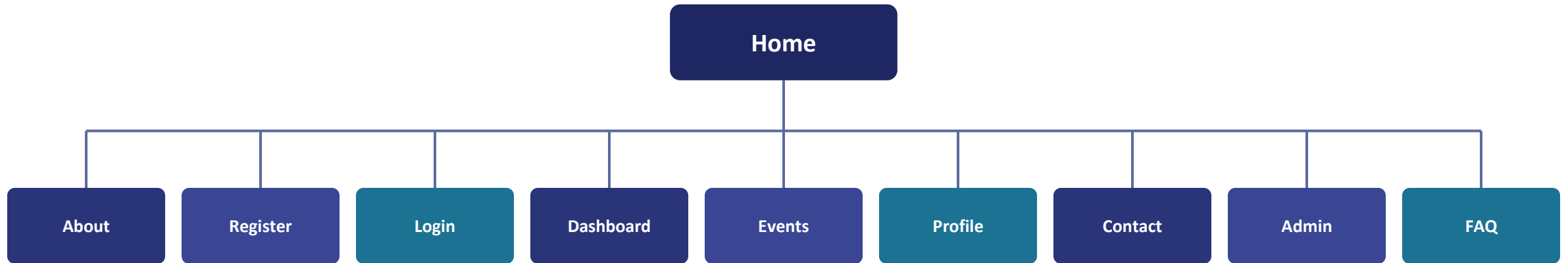
Fixes the minimum 10-page requirement early.

Aligns the team

Gives a shared reference before wireframing.

See the full StudentHub sitemap on the next slide.

Sitemap — StudentHub Portal

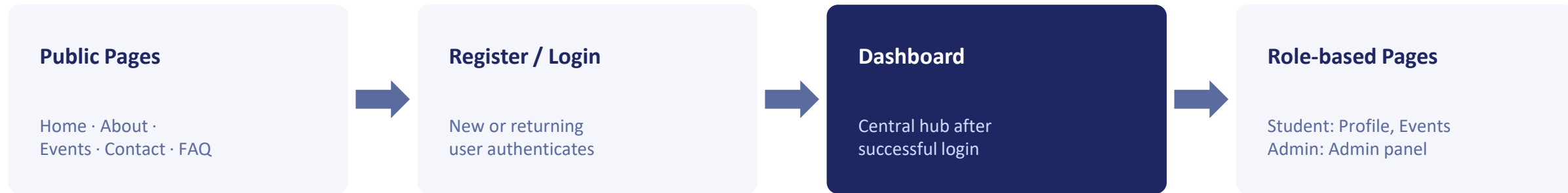


ACCESS LEGEND

- **Public** — Home, About, Register, Login, Events, Contact, FAQ — open to all visitors.
- **Student (authenticated)** — Dashboard, Profile, Events (registration) — require login.
- **Admin (role-restricted)** — Admin — manages students, faculty, events, notices and results.

Page Navigation Flow

Navigation is managed through a consistent top navigation bar on every page, plus role-based redirection after authentication.



Unauthorized access to Dashboard, Profile, or Admin redirects the user back to the Login page.

What is a Wireframe?

A wireframe is a low-fidelity, grayscale sketch of a page's layout. It shows where content blocks, navigation, and calls-to-action sit — without colors, fonts, or images — so structure can be validated before visual design.

- Focuses on layout and hierarchy, not visual style
- Fast and cheap to revise before coding begins
- Aligns stakeholders on placement of key elements
- Basis for the responsive CSS layout in Practical 3

FIDELITY SPECTRUM

Low-fi Wireframe

Boxes & labels — used here

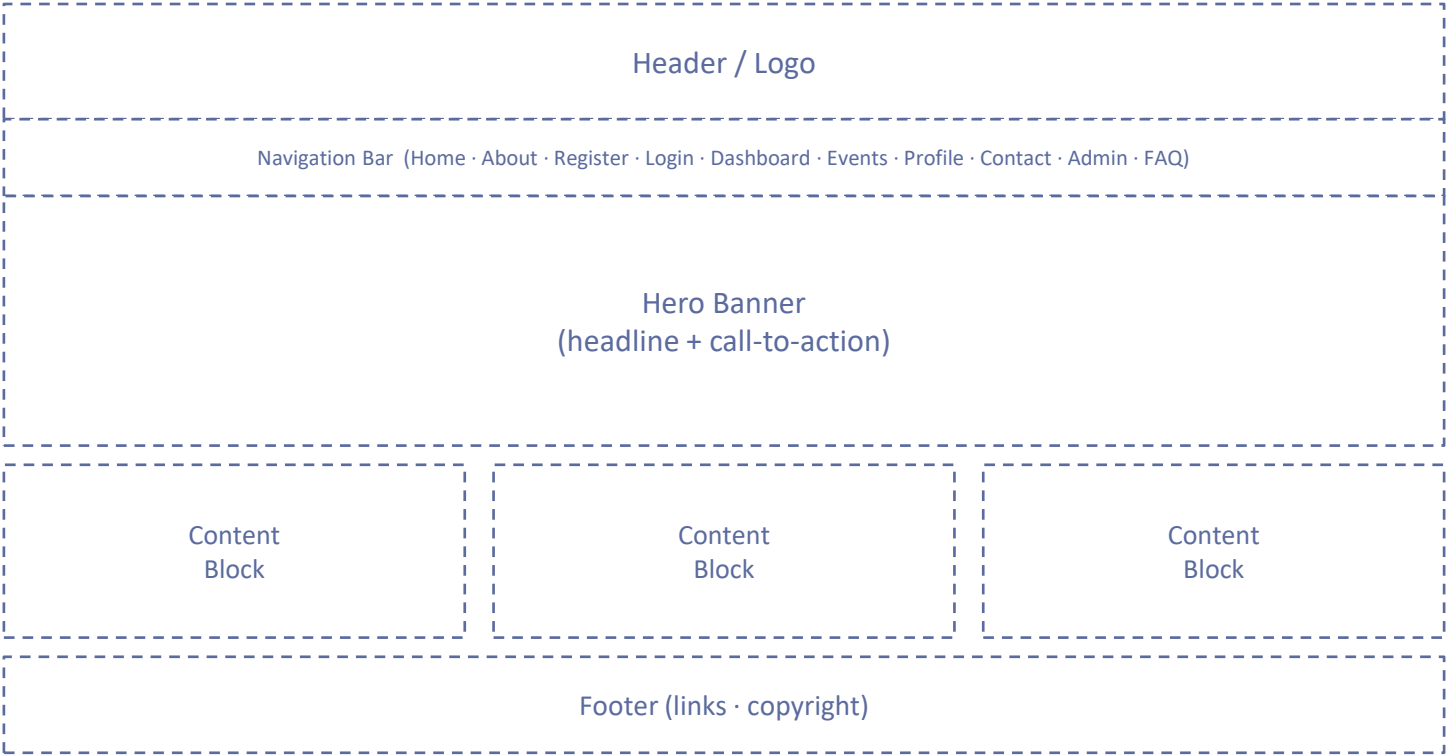
Mid-fi Mockup

Grayscale, real proportions

High-fi Prototype

Full color, interactive

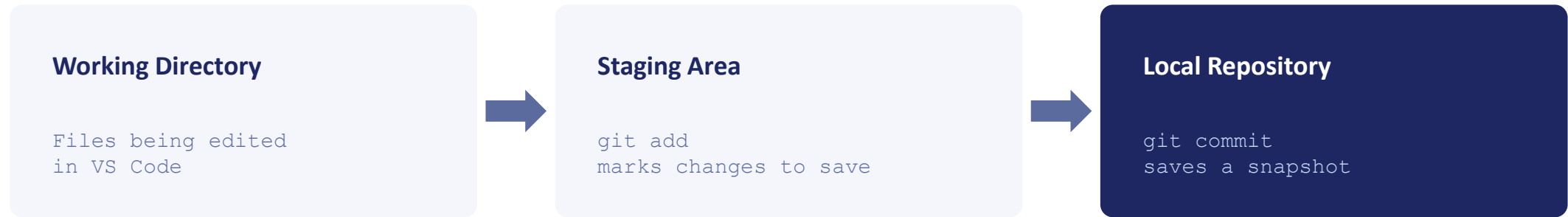
Low-Fidelity Wireframe — Home Page



Grayscale boxes label each zone by function — no color, imagery, or final copy at this stage.

Git — Version Control Basics

Git is a distributed version control system that records snapshots of a project over time, letting a developer track, compare, and roll back changes.



KEY COMMANDS

`git init``git status``git add .``git commit -m "msg"``git log``git branch`

GitHub — Repository Setup

GitHub hosts Git repositories in the cloud, giving the project a backup, a visible history, and a link that can be submitted for evaluation.

1

Create Repository

New repo "studenthub-portal" on GitHub, add a README and .gitignore.

2

Link Local Project

`git remote add origin <repo-url>`
connects the local folder to GitHub.

3

Push Initial Commit

`git push -u origin main` uploads the first snapshot of the project.

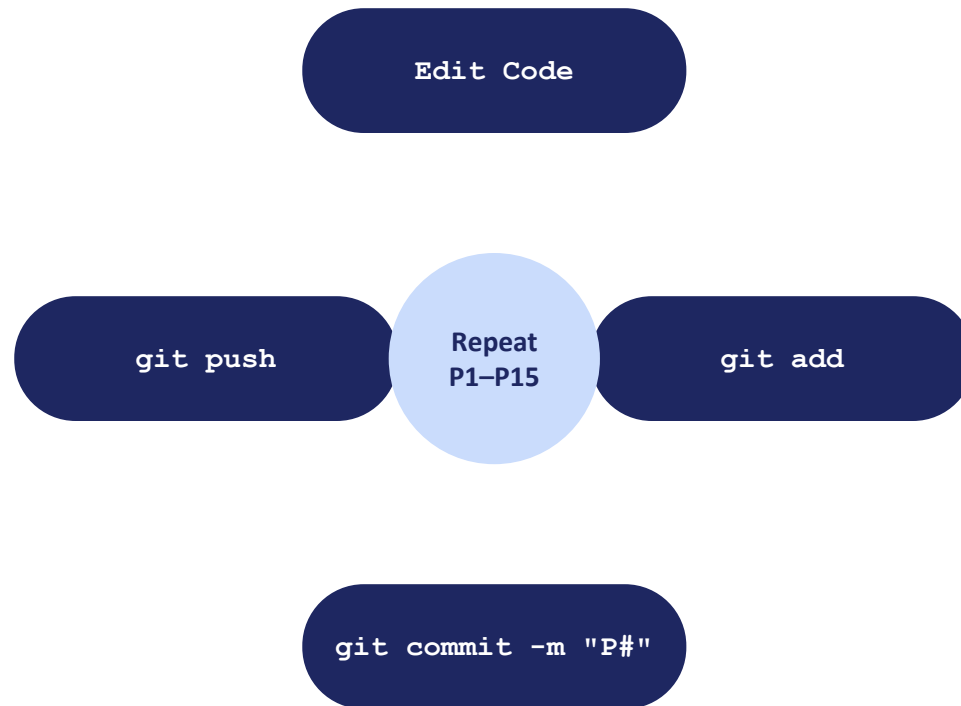
4

Submit Repository Link

Share the GitHub URL as part of the practical deliverables.

Maintaining GitHub Commits

The same edit–stage–commit–push cycle repeats after every practical, keeping a traceable history of the project's growth from P1 through P15.



COMMIT MESSAGE CONVENTION

P1: sitemap, wireframe, GitHub init

P2: semantic HTML5 pages added

P5: registration form validation

P11: student CRUD implemented

Project Folder Structure & README

```
studenthub-portal/  
├── index.html           (Home)  
├── about.html  
├── register.html  
├── login.html  
├── dashboard.html  
├── events.html  
├── profile.html  
├── contact.html  
├── admin.html  
├── faq.html  
├── /assets (css, js, images)  
├── /docs   (sitemap, wireframes)  
└── README.md
```

README.md SHOULD INCLUDE

- Project title & short description
- Problem statement and user roles
- Sitemap and wireframe references
- Setup / how to run the project
- Tech stack used
- Author & submission details

Learning Outcome

Students will be able to analyze requirements and prepare a structured project foundation — sitemap, wireframe, and version-controlled repository — before implementation begins.

EVALUATION CRITERIA

- Requirement clarity
- Sitemap completeness
- Role definition
- Repository structure
- Viva on project scope

EXTENSION

Intermediate

Create responsive wireframes in Figma.

Advanced

Prepare an ER diagram and REST-style route plan for future backend modules.